



PENRITH HIGH SCHOOL

**2015
HSC TRIAL EXAMINATION**

Mathematics Extension 1

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Answer each question on a new sheet of paper

Total marks–70

SECTION I

Pages 3–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

SECTION II

Pages 6–9

60 marks

- Attempt Questions 11–14
- Allow about 1 hours 45 minutes for this section

Student Number: _____

Teacher Name: _____

This paper MUST NOT be removed from the examination room

Assessor: T Bales

Section I

10 marks

Attempt Questions 1–10

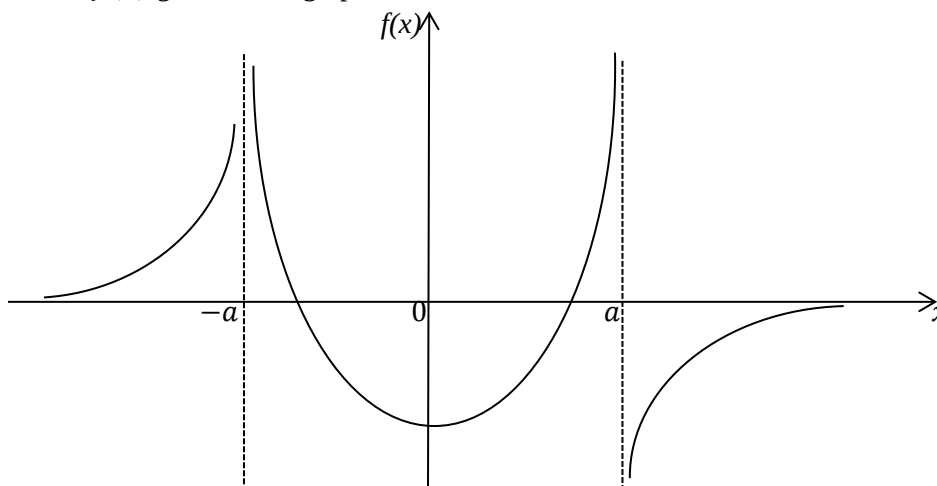
Allow about 15 minutes for this section

Use the provided multiple-choice answer sheet for Questions 1–10

1 For $x > 1$, $e^x - \ln x$ is:

- (A) $= 0$
- (B) > 0
- (C) < 0
- (D) $= e$

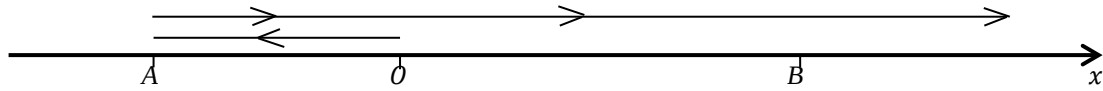
2 Consider the function $f(x)$ given in the graph below:



Which domain of the function $f(x)$ above is valid for the inverse function to exist?:

- (A) $x > 0$
 - (B) $-a < x < a$
 - (C) $0 < x < a$
 - (D) $x < 0$
- 3 What is the acute angle between the lines $2x - y - 7 = 0$ and $3x - 5y - 2 = 0$?
- (A) $4^\circ 24'$
 - (B) $32^\circ 28'$
 - (C) $57^\circ 32'$
 - (D) $85^\circ 36'$

- 4 A particle is moving in a straight line with velocity $v \text{ m/s}$ and acceleration $a \text{ m/s}^2$. Initially the particle started moving to the left of a fixed point O . The particle is noticed to be slowing down during the course of the motion from O to A . It turns around at A , keeps speeding up for the rest of the course of motion, passing O and B and continues. The particle never comes back. Take left to be the negative direction.



During the course of the particle's motion from O to A , which statement of the following is correct?

- (A) $v > 0$ and $a > 0$
- (B) $v > 0$ and $a < 0$
- (C) $v < 0$ and $a > 0$
- (D) $v < 0$ and $a < 0$
- 5 A particle is moving in a straight line with velocity, $v = \frac{1}{1+x} \text{ m/s}$ where x is the displacement of the particle from a fixed point O . If the particle was observed to have reached the position $x = -2 \text{ m}$ at a certain moment of time, then this particle:
- (A) will definitely reach the position $x = 1 \text{ m}$
- (B) may reach the position $x = 1 \text{ m}$
- (C) will never reach the position $x = 1 \text{ m}$
- (D) will come to rest before reaching $x = 1 \text{ m}$
- 6 If the rate of change of a function $y = f(x)$ at any point is proportional to the value of the function at that point then the function $y = f(x)$ is a:
- (A) Polynomial function
- (B) Trigonometric function
- (C) Exponential function
- (D) Quadratic function

7 Let α and β be any two acute angles such that $\alpha < \beta$. Which of the following statements is correct ?

- (A) $\sin \alpha < \sin \beta$
- (B) $\cos \alpha < \cos \beta$
- (C) $\operatorname{cosec} \alpha < \operatorname{cosec} \beta$
- (D) $\cot \alpha < \cot \beta$

8 Which of the following is a primitive function of $\sin^2 x + x^2$?

- (A) $x - \frac{1}{2} \sin 2x + \frac{x^3}{3} + c$
- (B) $\frac{1}{2}x - \frac{1}{4} \sin 2x + \frac{x^3}{3} + c$
- (C) $x - \frac{1}{2} \sin 2x + 2x + c$
- (D) $\frac{1}{2}x - \frac{1}{4} \sin 2x + 2x + c$

9 Consider the binomial expansion $(1+x)^n = 1 + n c_1 x + n c_2 x^2 + \cdots + n c_n x^n$. Which of the following expressions is correct?

- (A) ${}^n c_1 + 2 {}^n c_2 + \cdots + n {}^n c_n = n 2^{n-1}$
- (B) ${}^n c_1 + 2 {}^n c_2 + \cdots + n {}^n c_n = n 2^{n+1}$
- (C) ${}^n c_1 + {}^n c_2 + \cdots + {}^n c_n = 2^{n-1}$
- (D) ${}^n c_1 + {}^n c_2 + \cdots + {}^n c_n = 2^{n+1}$

10 Using the substitution, $u = 1 + \sqrt{x}$, find the value of $\int_1^4 \frac{1}{(1+\sqrt{x})^2} \frac{1}{\sqrt{x}} dx$ is:

- (A) $\frac{6}{5}$
- (B) $\frac{1}{3}$
- (C) $\frac{2}{3}$
- (D) $\frac{3}{2}$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Begin each question on a new sheet of paper. Extra sheets of paper are available.

In questions 11–14, your responses should include relevant mathematical reasoning and/or calculations

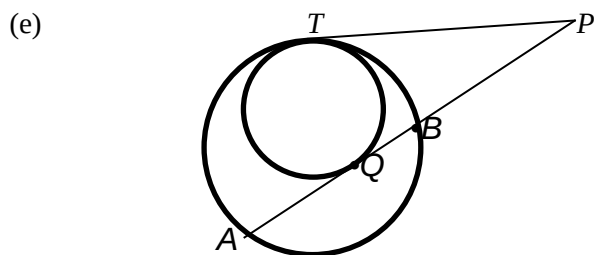
Question 11 (15 marks) Start a new sheet of paper.

(a) $(x + 1)$ and $(x - 2)$ are factors of $A(x) = x^3 - 4x^2 + x + 6$. Find the third factor. 2

(b) Find the coordinates of the point P which divides the interval AB internally in the ratio 2:3 with $A(-3, 7)$ and $B(15, -6)$ 3

(c) Solve the inequality $\frac{1}{4x-1} < 2$, graphing your solution on a number line. 3

(d) Use the method of mathematical induction to prove that, for all positive integers n : 3
$$1^2 + 3^2 + 5^2 + 7^2 + \cdots + (2n - 1)^2 = \frac{n}{3}(2n - 1)(2n + 1)$$



PT is the common tangent to the two circles which touch at T.

PA is the tangent to the smaller circle at Q, intersecting the larger circle at points B and A as shown.

i) State the property which would be used to explain why $PT^2 = PA \times PB$ 1

ii) If $PT = m$, $QA = n$ and $QB = r$, prove that $m = \frac{nr}{n - r}$ 3

Proceed to next page for question (12)

Question 12 (15 marks) Start a new sheet of paper.

- (a) The equation $\sin x = 1 - 2x$ has a root near $x = 0.3$. Use one application of Newton's methods to find a better approximation, giving your answer correct to 2 decimal places. 3
- (b) Five couples sit at a round table. How many different seating arrangements are possible if:
- i) there are no restrictions? 1
 - ii) each person sits next to their partner? 2
- (c) In the expansion of $(4 + 2x - 3x^2) \left(2 - \frac{x}{5}\right)^6$, find the coefficient of x^5 3
- (d) i) Write the binomial expansion for $(1 + x)^n$ 2
- ii) Using part (i), show that $\int_0^3 (1+x)^n dx = \sum_{k=0}^n \frac{1}{k+1} {}^nC_k 3^{k+1}$ 2
- iii) Hence show that $\sum_{k=0}^n \frac{1}{k+1} {}^nC_k 3^{k+1} = \frac{1}{n+1} [4^{n+1} - 1]$ 2

Proceed to next page for question (13)

Question 13 (15 marks) Start a new sheet of paper.

(a) Use the substitution $u = \frac{x}{\sqrt{1-x^2}}$ to show that $\frac{d}{dx} \left[\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) \right] = \frac{1}{\sqrt{1-x^2}}$ 3

(You can use the result that $\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + c$. You do not need to prove this).

(b) Give the exact value for $\int_{-\sqrt{3}}^3 \frac{dx}{3+x^2}$ 3

(c) The distinct points P, Q have parameters $t = t_1$ and $t = t_2$ respectively on the parabola $x = 2t, y = t^2$. The equations of the tangents to the parabola at P and Q respectively are given by:

$y - t_1x + t_1^2 = 0$ and $y - t_2x + t_2^2 = 0$ (You do not need to prove these)

i) Show that the equation of the chord PQ is $2y - (t_1 + t_2)x + 2t_1t_2 = 0$ 2

ii) Show that M , the point of intersection of the tangents to the parabola at P and Q , has coordinates $(t_1 + t_2, t_1t_2)$. 2

iii) α) Prove that for any value of t_1 , except $t_1 = 0$, there are exactly two values of t_2 for which M lies on the parabola $x^2 = -4y$. 3

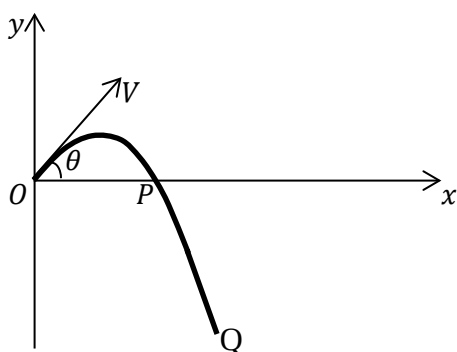
β) Find these two values of t_2 in terms of t_1 . 2

Proceed to next page for question (14)

Question 14 (15 marks) Start a new sheet of paper.

- (a) A particle P is moving in simple harmonic motion on the x axis, according to the law $x = 4\sin 3t$ where x is the displacement of P in centimetres from O at time t seconds.
- i) State the period and amplitude of the motion. 2
 - ii) Find the first time when the particle is 2cm to the positive side of the origin and its velocity at this time. 2
 - iii) Find the greatest speed and greatest acceleration of P 3

(b)



A projectile is fired from O , with speed $V\text{ms}^{-1}$, at an angle of elevation of θ to the horizontal. After t seconds, its horizontal and vertical displacements from O (as shown) are x metres and y metres, respectively.

- i) Prove that $x = Vt\cos\theta$ and $y = -\frac{1}{2}gt^2 + Vt\sin\theta$ 3
- ii) Show that the time taken to reach P is given by $t = \frac{2V\sin\theta}{g}$ 2
- iii) The projectile falls to Q , where its angle of depression from O is θ . 3
Prove that, in its flight from O to Q , P is the half-way point in terms of time.

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a \neq 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, x > 0$

Student Number: _____

Teacher Name: _____

Section I - Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

**Start
Here** →

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

a) $P(3) = 27 - 4 \times 9 + 3 + 6$
 $= 0$
 $\therefore (x-3)$ is third factor.

b) $A(-3, 7)$ $B(15, -6)$ $M:N$
 $2:3$

$$x = \frac{3(-3) + 2(15)}{5} = \frac{21}{5}$$

$$y = \frac{3(7) + 2(-6)}{5} = \frac{9}{5}$$

$$P\left(\frac{21}{5}, \frac{9}{5}\right)$$

c) $\frac{1}{4x-1} < 2$

$$4x-1 < 2(4x-1)^2$$

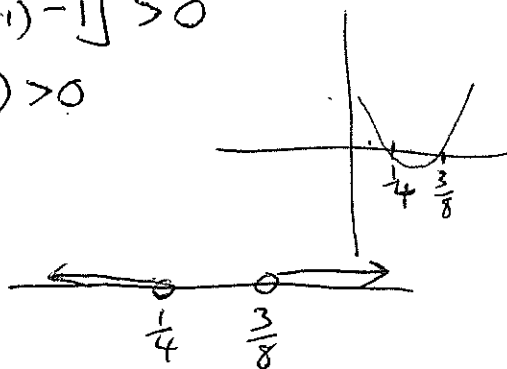
$$2(4x-1)^2 - (4x-1) > 0$$

$$(4x-1)[2(4x-1)-1] > 0$$

$$(4x-1)(8x-3) > 0$$

$$\therefore x < \frac{1}{4}$$

or ~~and~~ $x > \frac{3}{8}$



Mult. Choice

1) B 2) C 3) B 4) C 5) C

6) C 7) A 8) B 9) A 10) B

A lot of students found by polynomial division.

Extremely well done.
 problems occurred with getting signs wrong

students need to ensure a 'number line' is represented at the end and not shaded regions on a number plane

a) Test for $n=1$.

$$\text{L.H.S} = [2(1)-1]^2$$

$$= 1$$

$$\text{R.H.S} = \frac{1}{3}(1)(3)$$

$$= 1$$

$\therefore$ True for $n=1$

Assume true for $n=k$

$$1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{k}{3}(2k-1)(2k+1)$$

Test for $n=k+1$

that is we are required to prove

$$1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 = \frac{k+1}{3}(2k+1)(2k+3)$$

$$\text{L.H.S} = \frac{k}{3}(2k-1)(2k+1) + (2k+1)^2 \quad \text{by induction assumption}$$

$$= (2k+1) \left[\frac{k}{3}(2k-1) + (2k+1) \right]$$

$$= (2k+1) \left[\frac{2k^2 - k}{3} + (2k+1) \right]$$

$$= (2k+1) \left[\frac{2k^2 - k + 6k + 3}{3} \right]$$

$$= (2k+1) \left(\frac{1}{3} \right) (2k^2 + 5k + 3)$$

$$= \frac{1}{3}(2k+1)(k+1)(2k+3)$$

$$= \frac{k+1}{3}(2k+1)(2k+3)$$

$$= \text{R.H.S}$$

$\therefore$ proved by mathematical induction

Q(i) tangent-secant theorem.

(ii) $PT=m$ $QA=n$ $QB=r$ $PT^2 = PA \times PB$

$PT=PQ$ (equal tangents from an external point)

$$\therefore PQ = m$$

$$PA = PQ + QA$$

$$= m+n$$

$$PB = PQ - QB$$

$$= m-r$$

$$\therefore m^2 = (m+n)(m-r) \Rightarrow m^2 = m^2 + mn - rm - r^2$$

$$m(n-r) = r^2 \quad m = \frac{nr}{n-r}$$

well done,
some students
didn't
realise
 $S_{k+1} = S_k + T_{k+1}$
and
tried to prove

$$S_{k+1} = T_{k+1}$$

Only brief answer
was required.

students needed
to state
reason why
 $PT=PQ$

$$a) \sin x = 1 - 2x$$

$$f(x) : \sin x + 2x - 1 = 0$$

$$f'(x) : \cos x + 2$$

$$f(a) = a - \frac{f(a)}{f'(a)}$$

$$= 0.3 - \frac{\sin 0.3 + 0.6 - 1}{\cos 0.3 + 2}$$

$$= 0.33535$$

$$= 0.34 \text{ to 2 decimal places.}$$

$$b) i) 9!$$

$$(ii) 4! \times 2^5$$

$$c) (4 + 2x - 3x^2)(2 - \frac{x}{5})^6$$

$$\text{General term: } {}^6C_k 2^{6-k} (-1)^k (\frac{x}{5})^5$$

$$\text{for } (2 - \frac{x}{5})^6$$

$${}^6C_k 2^{6-k} 5^{-k} (-1)^k x^k$$

when $k=5$. multiply by 4 (other brackets)

$$- {}^6C_5 2 \cdot 5^{-5} (4) = \boxed{\frac{-48}{3125}}$$

when $k=4$ multiply by 2 (other brackets)

$${}^6C_4 2^2 5^{-4} (2) = \boxed{\frac{120}{625}}$$

when $k=3$ multiply by (-3) (other brackets)

$$- {}^6C_3 2^3 5^{-3} (-3) = \boxed{\frac{480}{125}}$$

$$\frac{-48}{3125} + \frac{120}{625} + \frac{480}{125} = \frac{12552}{3125} \text{ or } 4.01664$$

some students forgot to rearrange to $=0$.

some student put equation around the wrong way so that the numerator and denominator gave a negative overall value.

students forgot that all 5 pairs could be swapped between the pair.

General problems with signs.

Suggested Solutions

Marker's Comments

$$d(i) (1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n.$$

well done

$$\begin{aligned} (ii) \int_0^3 (1+x)^n dx &= \int_0^3 \left(\binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n \right) dx \\ &= \left[\binom{n}{0}x + \binom{n}{1}\frac{x^2}{2} + \binom{n}{2}\frac{x^3}{3} + \dots + \binom{n}{n}\frac{x^{n+1}}{n+1} \right]_0^3 \\ &= \left(3\binom{n}{0} + \binom{n}{1}\frac{3^2}{2} + \binom{n}{2}\frac{3^3}{3} + \dots + \binom{n}{n}\frac{3^{n+1}}{n+1} \right) - 0 \\ &= \sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} 3^{k+1} \end{aligned}$$

students needed to demonstrate integration

$$\begin{aligned} (iii) \int_0^3 (1+x)^n dx &= \left[\frac{(1+x)^{n+1}}{n+1} \right]_0^3 \\ &= \frac{4^{n+1}}{n+1} - \frac{1}{n+1} = \frac{1}{n+1} [4^{n+1} - 1] \end{aligned}$$

if students realised method they were successful.

$$\therefore \text{from (ii) (iii)} \quad \sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} 3^{k+1} = \frac{1}{n+1} [4^{n+1} - 1]$$

a) Let $u = \frac{x}{\sqrt{1-x^2}}$ Let $y = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$

$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}}$$

$$\frac{dy}{du} = \frac{d}{du}(\tan^{-1}u) = \frac{1}{1+u^2}$$

$$\frac{dy}{du} = \frac{1}{1 + \frac{x^2}{1-x^2}} = \frac{1}{\frac{1-x^2+x^2}{1-x^2}} = \underline{\underline{\frac{1-x^2}{1-x^2}}}$$

$$\begin{aligned} \frac{du}{dx} &= \frac{d}{dx}(x(1-x^2)^{-\frac{1}{2}}) \\ &= x \times \frac{1}{2}(1-x^2)^{-\frac{3}{2}} \times -2x + 1(1-x^2)^{-\frac{1}{2}} \\ &= \frac{-x^2}{(\sqrt{1-x^2})^3} + \frac{1}{\sqrt{1-x^2}} = \underline{\underline{\frac{-x^2 + 1 - x^2}{(\sqrt{1-x^2})^3}}} \end{aligned}$$

$$\frac{dy}{dx} = (1-x^2) \times \frac{1}{(\sqrt{1-x^2})^3} = \boxed{\frac{1}{\sqrt{1-x^2}}}$$

b) $\int_{-\sqrt{3}}^{\sqrt{3}} \frac{1}{3+x^2} dx = \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{x}{\sqrt{3}} \right]_{-\sqrt{3}}^{\sqrt{3}}$

$$= \frac{1}{\sqrt{3}} \left(\tan^{-1} \frac{\sqrt{3}}{\sqrt{3}} - \tan^{-1} \frac{-\sqrt{3}}{\sqrt{3}} \right)$$

$$= \frac{1}{\sqrt{3}} \left(\tan^{-1} \sqrt{3} - \tan^{-1} (-1) \right)$$

$$= \frac{1}{\sqrt{3}} \left(\frac{\pi}{3} - -\frac{\pi}{4} \right)$$

$$= \underline{\underline{\frac{7\pi}{12\sqrt{3}} \quad \left(\text{or } \frac{7\pi\sqrt{3}}{36} \right)}}$$

Some students did not use the given substitution which was required.

Some students did not seem familiar with $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

Note: If you use Product rule $u=x, v=(1-x^2)^{-\frac{1}{2}}$
For quotient rule $u=x, v=(1-x^2)^{\frac{1}{2}}$
Students were mixing these up.

This question was well answered.

Some students did not know the range of $\tan^{-1}$ to be $-\frac{\pi}{2}$ to $\frac{\pi}{2}$

Suggested Solutions

Marker's Comments

$$13c i. m = \frac{t_2^2 - t_1^2}{2t_2 - 2t_1} = \frac{(t_2 + t_1)(t_2 - t_1)}{2(t_2 - t_1)} = \frac{t_2 + t_1}{2}$$

$$y - t_1^2 = \frac{(t_1 + t_2)}{2}(x - 2t_1)$$

$$2y - 2t_1^2 = (t_1 + t_2)x - 2t_1^2 - 2t_1t_2$$

$$2y - (t_1 + t_2)x + 2t_1t_2 = 0$$

$$ii. y = t_1x - t_1^2$$

$$y = t_2x - t_2^2$$

$$t_1x - t_1^2 = t_2x - t_2^2$$

$$t_1x - t_2x = t_1^2 - t_2^2$$

$$x(t_1 - t_2) = (t_1 + t_2)(t_1 - t_2)$$

$$x = t_1 + t_2$$

$$\text{sub into } y = t_1x - t_1^2$$

$$y = t_1(t_1 + t_2) - t_1^2$$

$$= t_1^2 + t_1t_2 - t_1^2$$

$$y = t_1t_2$$

$$M(t_1 + t_2, t_1t_2)$$

$$iii. x^2 = -4y$$

$$(t_1 + t_2)^2 = -4(t_1t_2)$$

$$t_1^2 + 2t_1t_2 + t_2^2 = -4t_1t_2$$

$$t_1^2 + 6t_1t_2 + t_2^2 = 0$$

$$\Delta = 36t_1^2 - 4 \cdot 1 \cdot t_1^2$$

$$= 32t_1^2$$

$$\Delta > 0 \text{ for all } t_1 \therefore 2 \text{ values for all } t_2$$

• Some students didn't know where to start.

• sub M into $x^2 = -4y$

• use discriminant to show 2 values ie $\Delta > 0$

$$t_2 = \frac{-6t_1 \pm \sqrt{32t_1^2}}{2}$$

$$= \frac{-6t_1 \pm 4\sqrt{2}t_1}{2}$$

$$= -3t_1 \pm 2\sqrt{2}t_1$$

$$= (-3 \pm 2\sqrt{2})t_1$$

Some students
found t_1
in terms of t_2

i) $x = 4\sin 3t$

amplitude = 4

period = $\frac{2\pi}{n} = \frac{2\pi}{3}$

ii) $2 = 4\sin 3t$

$\frac{1}{2} = \sin 3t$

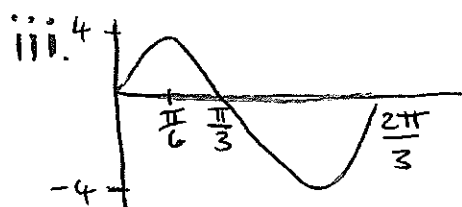
$3t = \frac{\pi}{6}$

$t = \frac{\pi}{18}$

$\dot{x} = 12\cos 3t$

$= 12\cos 3\left(\frac{\pi}{18}\right)$

$= 6\sqrt{3}$



max velocity when $t=0$

$\dot{x} = 12\cos 3(0)$

$= 12\text{ cm s}^{-1}$

max acceleration when $t = \frac{\pi}{2}$
(positive direction)

$\ddot{x} = -36\sin 3t$

$= 36\text{ cm s}^{-2}$

• errors were made differentiating $x = 4\sin 3t$

• common error

$t = \frac{\pi}{18} = 10\text{ sec}$

• errors differentiating $\dot{x} = 12\cos 3t$

• using degrees instead of radians

• -36 cm s^{-2} was accepted

Suggested Solutions

Marker's Comments

$$\text{bi. } \ddot{x} = 0$$

$$\dot{x} = c_1$$

$$\text{when } t=0 \dot{x} = v \cos \theta$$

$$\therefore c_1 = v \cos \theta$$

$$\dot{x} = v \cos \theta$$

$$x = vt \cos \theta + c_2$$

$$\text{when } t=0 \ x=0 \therefore c_2=0$$

$$x = vt \cos \theta$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + c_3$$

$$\text{when } t=0 \dot{y} = v \sin \theta$$

$$\therefore c_3 = v \sin \theta$$

$$\dot{y} = -gt + v \sin \theta$$

$$y = -\frac{gt^2}{2} + vt \sin \theta + c_4$$

$$\text{when } t=0 \ y=0 \therefore c_4=0$$

$$y = -\frac{gt^2}{2} + vt \sin \theta$$

• errors were common when evaluating constants

$$\text{bii. } P \rightarrow y=0$$

$$0 = -\frac{1}{2}gt^2 + vt \sin \theta$$

$$0 = t(v \sin \theta - \frac{gt}{2})$$

$$t=0 \text{ or } v \sin \theta - \frac{gt}{2} = 0$$

$$\frac{gt}{2} = v \sin \theta$$

$$t = \frac{2v \sin \theta}{g}$$

$$\text{iii. } \tan \theta = -\frac{y}{x}$$

$$\tan \theta = -\frac{(vt \sin \theta - \frac{1}{2}gt^2)}{vt \cos \theta}$$

$$\tan \theta = -\tan \theta + \frac{gt}{2v \cos \theta}$$

$$2 \tan \theta = \frac{gt}{2v \cos \theta}$$

$$\frac{4v \sin \theta}{g} = t \text{ which is twice } \frac{2v \sin \theta}{g}$$

• be careful quoting the cartesian eq'n for the trajectory (a method some) used
You may need to derive it.